Benchmarking Deep Learning Models for Non-Coding Variant Prediction

Team 13

Kavya Puranam
kpura3@illinois.edu

Ellen Yang
yaweny2@illinois.edu

Abstract

Non-coding regions govern when, where, and how genes are expressed. Understanding how variants within these elements affect gene regulation is crucial for enhancing our ability to predict disease risk and identify new therapeutic targets. This study evaluates whether modern deep learning models outperform traditional annotation-based approaches across key genomic contexts such as enhancers, promoters, and untranslated regions. Model performance is assessed using three methods, including CADD (Combined Annotation Dependent Depletion) (Rentzsch et al., 2019), a Classical ensemble model, DeepSEA, a Deep learning CNN for regulatory variant prediction (Zhou & Troyanskaya, 2015), and Enformer, a transformer-based deep learning model that captures long-range regulatory effects (Avsec et al., 2021). Our results show that CADD achieves the most balanced and robust performance across all evaluation metrics, reliably separating pathogenic from benign variants. Enformer achieved the strongest performance in ranking pathogenic variants, reflected by the high AUROC, but limited by low recall under small and imbalanced test conditions. DeepSEA demonstrated weaker discriminatory performance for pathogenic variants despite evaluation on a larger dataset. Together, these findings highlight the continued reliability of annotation-based models while showing the promise of transformer-based architectures for regulatory variant prioritization as larger, well-processed, and diverse datasets become available. By comparing the performance of novel biologically informed deep learning models on regulatory variants, this work aims to address cases that remain unresolved by solely coding-region expression approaches.

1 Introduction

Variants consist of genetic changes to DNA sequences as compared to the reference genome, typically the Human Reference Genome. Variants are caused by mutations, which can either be beneficial or deleterious. Single Nucleotide Polymorphisms (SNPs) are the most common genetic variation in the human genome, simply a switch between a single base pair, but they can cause major genetic changes. Another variation is an insertion/deletion mutation (indels), which removes or adds a base pair to the DNA sequence. The consequences of these mutations differ in severity as well. Silent mutations do not cause changes to the protein sequence, allowing it to be coded as normal (Torgerson & Ochs, 2014). Missense mutations occur when a SNP changes the amino acid being coded for, changing the functionality of the protein itself, while Nonsense mutations prematurely create a stop codon, ending protein synthesis early. Frameshift mutations can occur due to indels leading to incorrect reads of amino acids downstream from the insertion or deletion, or reaching a stop codon prematurely, preventing the production of the protein (Gorlov et al., 2018). While many of these mutations affect protein synthesis, these mutations also occur in the non-coding region of the genome, causing other unintended effects.

The non-coding region is the large majority of the human genome, totaling 98.5% of the human genome. While they do not encode proteins, they play a crucial role in regulating gene expression. They control gene expression through different regulatory elements. These include promoters that control transcription initiation, enhancers that activate gene expression over long genomic distances, and untranslated regions (UTRs) that influence mRNA stability and translation efficiency (Smith et al., 2012). Deleterious mutations in non-coding regions can disrupt gene regulation rather than changing the protein itself. Cis-regulatory variants can change gene expression in three main ways, including breaking a transcription factor binding site, lowering gene expression, creating a new transcription factor binding site, introducing new regulatory activity, and finally disrupting CTCF binding, breaking TAD boundaries, allowing for new promoter-enhancer interactions that alter expression. With protein synthesis, it is easier to find causal mutations; however, mutations altering gene expression have a widespread cascading effect that is complex and difficult to detect.

By focusing on regulatory variants in human tissues, we can deepen our understanding of the genetic mechanisms underlying diseases and inform genetic disorders. Using computational methods can rapidly increase the insights we can gain from the non-coding region. In this study, we evaluate which deep learning models are best at predicting the regulatory impact of non-coding variants across different genomic contexts, such as enhancers, promoters, and untranslated regions. We are also interested in assessing how well modern deep learning models, such as transformers and Convolutional Neural Networks (CNNs), perform on non-coding regulatory variants over traditional annotation-based models. We benchmark three representative approaches to non-coding variant prediction: CADD, a classical annotation-based ensemble model; DeepSEA, a CNN trained on regulatory sequence features; and Enformer, a transformer-based model that captures long-range regulatory interactions. While many deep learning models evaluate mutation effects in coding regions, there is a lack of rigorous work on how variants in promoters, enhancers, and other non-coding elements can alter gene expression. Our project aims to fill this gap by focusing on the non-coding region and prioritizing regulatory variants that may resolve cases left unsolved by coding-only expression. Ultimately, we seek to show how non-coding region mutation can guide diagnostics of disease.

2 Related Work

We base our study on the existing approaches to non-coding variants using deep learning models established for that specific task. The models that we are using, Enformer and DeepSEA, are landmark papers in predicting the functional impact of SNPs using sequence-to-expression models. DeepSEA's key findings were to be the first model to predict the chromatin effect of a variant from a DNA sequence directly, which proved that models can prioritize functional variants based on the change in predicted chromatin structure (Zhou &

Troyanskaya, 2015). Enformer's key findings were that it improved the accuracy of variant effect prediction on gene expression by modeling enhancer-promoter interactions (Avsec et al., 2021). Similarly, other models such as Basenji and Borzoi use CNN and transformer architecture to use quantitative genomic profiles such as CHI-seq across the genome to predict gene expression (Kelley et al., 2018) and improve causal variant identification and mRNA level predictions (Linder et al., 2023) .

Despite these advances, these models face limitations in their application in clinical settings. Benchmarking these models does not account for how well models perform on rare variants or indels compared to common SNPs. There is a lack of information on benchmarks for cell-type-specific causal variants. Additionally, most benchmarks focus on models being able to predict changes due to genetic variants, but there is no standardized framework as of now to benchmark whether a model is correctly identifying a motif disruption or if it is finding statistical patterns in the data itself. Overall, there is limited evaluation of head-to-head benchmarking of models and across specific regulatory contexts such as promoters, enhancers, and UTRs, which is where what we aim to address with this study.

3 Methods

Overview of Benchmarking Framework and Evaluation Strategy

Our benchmarking framework is designed to evaluate the performance of distinct modeling paradigms for non-coding variants prediction under a unified and consistent evaluation setting. The three models we selected represent key stages in the evolution of genomic prediction models: CADD, DeepSEA, and Enformer. The three models cover three major methodological classes, an annotation-based ensemble method, a convolutional neural network (CNN) framework, and a transformer-based sequence model. While CADD serves as a classical, foundational ensemble model, DeepSEA and Enformer represent more advanced frameworks utilizing recent deep neural network architectures. Despite these fundamental architectural differences, all three models share the common objective of predicting genomic variant scores that reflect potential deleteriousness. This shared purpose provides a meaningful basis for a comparison of their predictive power for non-coding variants.

Beyond architectural differences, these models exhibit varying requirements regarding the input genome assembly representation. CADD and Enformer natively support the hg38 reference genome, which is the most accurately sequenced human genome. However, DeepSEA was originally developed and trained using hg19, an earlier version of the human genome reference sequence. To ensure consistency and comparability across all three models, we performed the necessary data processing and creation of a consistent input dataset. This step ensures that all models are evaluated using the same reference coordinates.

Data Sources and Variant Selection

We planned to use a Variant Call Format text file containing the chromosome, position, reference allele, and alternate allele for each variant as the input data for each model. The data source used was the ncVarDB pre-annotated list of non-coding human genome variants with benign controls. The dataset contained 721 pathogenic variants in a VCF file and 7228 benign variants. This database compiled and published non-coding variants with proven evidence of functional consequences. As this database contained well-documented and curated annotated VCF data, we decided it would be best to use in our evaluation of the models' performances. After we found our datasets, we needed to adapt the VCF for each of the input requirements for each model. Since the ncVarDB dataset also had a script for

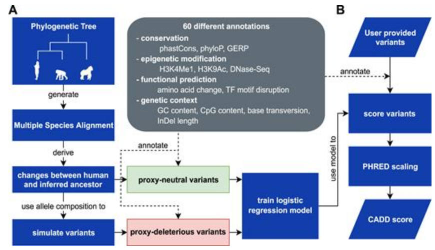

Figure 1: Overview of CADD framework (Rentzsch et al., 2019)

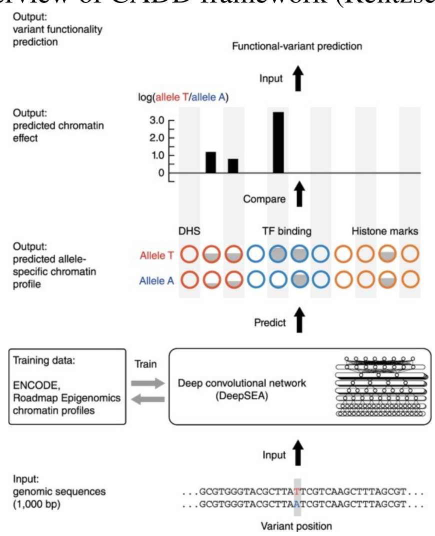

Figure 2: Overview of DeepSEA framework (Zhou & Troyanskaya, 2015)

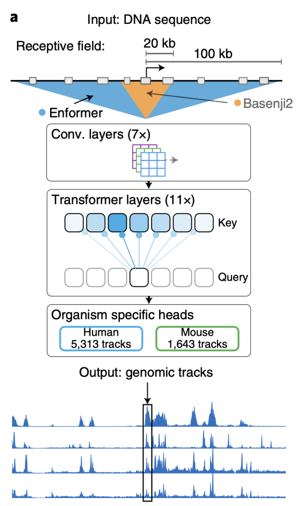

Figure 3: Overview of Enformer transformers-based framework (Avsec et al., 2021)

CADD we used that as the annotated regulatory variant input. For DeepSEA we created a 1000bp window for variant data in ncVarDB, including both pathogenic and benign variants. Because DeepSEA model is trained exclusively on the hg19 human reference genome, variants originally provided in hg38 were then converted to hg19 prior to data processing steps. For Enformer, we created a window of the maximum sequence length the Enformer model takes, which is 393,216 bp then we skipped variants near chromosome boundaries and edges to avoid redundancy, misaligned variants, and checked against the reference hg38. The same method was used for the benign data preparation as well. Since there were very few variants for the pathogenic data, we took a random sample of the same number of variants for the benign data.

Model Implementation and Inference
For all three models examined in our project, we did not attempt to reimplement or retrain the original frameworks. Instead, we relied on publicly available, officially released pretrained models and inference pipelines to obtain non-coding variant effect scores. This approach ensures that each model is evaluated in its original settings and can achieve its best results, thereby avoiding performance differences introduced by reimplementation and retraining choices. All models were treated as black-box predictors, and the benchmarking was evaluated solely on their performance output scores.

Due to the difference in architecture, input requirements, and supported human genome assembly among models, variant scores are obtained using model-specific approaches.

CADD We used the official CADD scoring framework provided through its website and offline pipeline. The most recent CADD release supports the GRCh38(the hg38 human reference genome) and provides precomputed scores. We used CADD as the classical baseline for our comparison. However, due to the annotation-heavy nature of CADD, the scoring process was computationally slower than that of the other models.

DeepSEA We used the most recent Sei[1] variant-effect prediction model, which extends the original DeepSEA framework. The model accepts variants in VCF format but was trained on hg19 reference genome. Therefore, all input variants were converted and restricted to hg19 coordinates.

Enformer We applied the publicly released pretrained transformer model and official source code provided by the authors. The setup allows us to evaluate Enformer in its intended configuration and obtain the scoring results.

Evaluation Metrics Development
To conduct the benchmarking, we first obtained and normalized the scoring results from each model. Since each model outputs scores in a different format, we generalized them to fit into a uniform scoring context to serve as the reference for the benchmark. The following scores were used by each model to represent the predictions: For CADD, we used the PHRED-scaled score. For DeepSEA, we used the sequence class max absolute difference. For Enformer, we used the SNP Activity Difference (SAD) score. All models are capable of predicting deleterious variants using these respective scores.

After gathering the results scores, we performed the benchmark. To evaluate and compare each model's performance, we utilized both discrimination and threshold-dependent metrics. We computed the Area Under the Receiver Operating Characteristic curve (AUROC) and the Area Under the Precision–Recall Curve (AUPRC) to measure the model's ability to distinguish between pathogenic and benign variants. We also computed accuracy, precision, recall, and F1 score to evaluate the overall predictive correctness for each model at a fixed threshold. Together, these metrics allow us to rate the models' effectiveness in correctly identifying and predicting deleterious variants.

Challenges
We faced issues with the data quality, such as mismatches with the reference genome or

[1] https://humanbase.readthedocs.io/en/latest/sei.html

genome inconsistencies, which disrupted computation and biased performance. In order to mitigate this, we checked each input sequence against the reference and fixed or dropped discordant sites while keeping true biological variants unchanged.

4 Results

Due to the high computational cost for both Enformer and CADD models, we were unable to process the entire non-coding variant dataset for all three models. To address this limitation, we focused on critical regions of interest, skipping variants near chromosome edges, which are less informative, to prioritize regions crucial to mutation analysis. In contrast, the relatively lightweight DeepSEA framework allowed us to utilize the complete ncVar dataset, which includes 721 pathogenic variants and 7,228 benign variants. For Enformer, we restricted the analysis due to computational constraints to a small but balanced subset of 21 pathogenic and 21 benign variants that are crucial to detecting deleterious variants. While we successfully obtained scores for all 721 pathogenic variants for CADD, generating scores for the full benign set proved impractical within the available time and resources. To ensure a consistent and fair comparison across models, we therefore evaluated CADD using the same benign subset used for Enformer.

The benchmarking results are summarized in Table 1. The benchmark scores shows that CADD and Enformer both achieve high AUROC values, over 0.95 for both models, with Enformer scoring 0.96, the highest among the three models.

Table 1: Benchmark Scores

Model	AUROC	AUPRC	Accuracy	Precision	Recall	F1
CADD	0.95215	0.99842	0.94992	0.97045	0.97806	0.97424
DeepSEA	0.72867	0.19651	0.80165	0.18956	0.43328	0.26374
Enformer	0.96371	0.97024	0.523809	1.0	0.04676	0.09090

The metric scores for CADD appear to be the most stable, with all values above 0.9, reflecting the highest overall performance across multiple metrics. While the DeepSEA model exhibits strong accuracy, its comparatively low precision and F1 score indicate a lack of reliability in accurately identifying and localizing deleterious variants. For Enformer, both the AUROC and AUPRC scores are notably high, both exceeding 0.9, and a perfect precision score. However, the extremely low recall and subsequent low F1 score are directly attributable to the limitation imposed by the small variant subset used for its evaluation.

5 Discussion

CADD demonstrates the most balanced and robust performance across all metrics, achieving high scores in every metric we measured. Even with evaluation restricted to only 21 benign variants, it consistently separates pathogenic from benign variants, aligning with its design as a genome-wide deleteriousness prioritization model.

Enformer outperforms at pathogenic identification, as evidenced by its highest AUROC score. However, its extremely low recall and low overall accuracy suggest that the small-sample evaluation reflects a limitation in the current test set rather than an inherent performance flaw.

Although DeepSEA benefits from full-dataset evaluation, it shows a weaker ability to distinguish between pathogenic and benign variants compared to the other models. While its accuracy is relatively high due to the large benign set, its low AUPRC, precision, and F1 scores reflect a lack of discriminatory ability in distinguishing pathogenic variants from data noise.

The benchmarking results generally reflect the design principles of each model framework. Enformer, as a transformer-based model capable of modeling long-range genomic dependencies, shows strong potential in ranking deleterious variants. At the same time, the stable performance of the classical ensemble model CADD highlights the enduring effectiveness of annotation-based integration, reinforcing its role as a reliable baseline in both research and clinical contexts.

6 Conclusion

To further the work in this field of regulatory variant predictions, we can explore other deep learning models for benchmarking. EPInformer (2024) is a scalable deep learning model that predicts gene expression through promoter-enhancer sequences and interactions. ExPecto (2019) is a tissue-specific gene expression prediction from sequences; however it doesn't use variant information. The impact of this research is broad, uncovering methods to help human health that have never been found before. Specifically, deep learning models are uniquely positioned to tackle biological problems never encountered before. Predictive modeling is able to conquer the large-scale non-coding region data to filter variants for the most significant regulatory activity. It is also able to capture non-linear patterns, especially with gene regulation and transposable elements, such as the unpredictable and complex dynamics of insertions and gene functions. Overall, our benchmarking results demonstrate that classical and modern deep learning approaches each offer distinct strengths in non-coding variant prediction. CADD showed the most balanced and robust performance across all evaluated metrics, consistently distinguishing pathogenic from benign variants even under limited sample conditions, demonstrating its reliability as a genome-wide deleteriousness prioritization tool. Enformer achieved the strongest performance in pathogenic variant ranking, as reflected by its superior AUROC, highlighting the promise of transformer-based architectures for modeling long-range regulatory dependencies. However, its low recall and accuracy in this study emphasize the sensitivity of deep learning models to dataset size and class imbalance rather than inherent model limitations. DeepSEA, while benefiting from full-dataset evaluation, showed reduced discriminatory power for pathogenic variants, suggesting that convolutional architectures may struggle to capture complex regulatory signals in highly imbalanced non-coding datasets. Collectively, these findings reinforce the continued relevance of annotation-based models as robust baselines, while also demonstrating the potential of advanced deep learning methods to improve regulatory variant prioritization as datasets grow in size and diversity.

7 Team Work Division

The following figure describes the team's tasks and the dates at which each task was completed. The person assigned to complete the task is listed; however, much of the technical work was done together during co-working sessions. float

Task	Description	Date	Assigned
Review Literature	Survey 3–5 papers on non-coding variant prediction	Oct 31	Ellen Yang & Kavya Puranam
Submit Preproposal	1–2 page proposal	Oct 31	Ellen Yang & Kavya Puranam
Data Exploration and Cleaning	Curated non-coding genomic data with available datasets	Nov 5	Kavya Puranam
Dataset Creation	Assemble available non-coding genomic data using ncVarDB	Nov 7	Kavya Puranam
1st Model Architecture Setup	Implement CADD and run model with datasets	Nov 26	Ellen Yang
2nd Model Architecture Setup	Implement DeepSEA and run model with datasets	Nov 27	Ellen Yang
3rd Model Architecture Setup	Implement Enformer and run model with datasets	Nov 28	Ellen Yang and Kavya Puranam
In-Class Presentation	Prepare 5–10 minute presentation	Dec 12	Ellen Yang & Kavya Puranam
Final Report	8–10 page report excluding references, NeurIPS Overleaf template	Dec 16	Ellen Yang & Kavya Puranam

References

[1] Richard Boland C. (2017). Non-coding RNA: It's Not Junk. Digestive diseases and sciences, 62(5), 1107–1109. https://doi.org/10.1007/s10620-017-4506-1

[2] Rentzsch, P., Witten, D., Cooper, G. M., Shendure, J., & Kircher, M. (2019). CADD: predicting the deleteriousness of variants throughout the human genome. Nucleic acids research, 47(D1), D886–D894. https://doi.org/10.1093/nar/gky1016

[3] Zhou, J., & Troyanskaya, O. G. (2015). Predicting effects of noncoding variants with deep learning–based sequence model. Nature Methods, 12(10), 931–934. https://doi.org/10.1038/nmeth.3547

[4] Avsec, Ž., Agarwal, V., Visentin, D., Ledsam, J. R., Grabska-Barwinska, A., Taylor, K. R., Assael, Y., Jumper, J., Kohli, P., & Kelley, D. R. (2021). Effective gene expression prediction from sequence by integrating long-range interactions. Nature Methods, 18(10), 1196–1203. https://doi.org/10.1038/s41592-021-01252-x

[5] Steyerberg, E. W., Vickers, A. J., Cook, N. R., Gerds, T., Gonen, M., Obuchowski, N., Pencina, M. J., & Kattan, M. W. (2010). Assessing the performance of prediction models: a framework for traditional and novel measures. Epidemiology (Cambridge, Mass.), 21(1), 128–138. https://doi.org/10.1097/EDE.0b013e3181c30fb2

[6] Linder, J., Srivastava, D., Yuan, H., Agarwal, V., & Kelley, D. R. (2025). Predicting RNA-seq coverage from DNA sequence as a unifying model of gene regulation. *Nature Genetics 2025 57:4*, *57*(4), 949–961. https://doi.org/10.1038/s41588-024-02053-6

[7] Kelley, D. R., Reshef, Y. A., Bileschi, M., Belanger, D., McLean, C. Y., & Snoek, J. (2018). Sequential regulatory activity prediction across chromosomes with convolutional neural networks. *Genome Research*, *28*(5), 739–750. https://doi.org/10.1101/GR.227819.117/-/DC1

[8] Biggs, H., Parthasarathy, P., Gavryushkina, A., & Gardner, P. P. (2020). ncVarDB: a manually curated database for pathogenic non-coding variants and benign controls. *Database*, *2020*. https://doi.org/10.1093/DATABASE/BAAA105

[9] Torgerson, T., & Ochs, H. (2014). Genetics of Primary Immune Deficiencies. *Stiehm's Immune Deficiencies*, 73–81. https://doi.org/10.1016/B978-0-12-405546-9.00003-0

[10] Gorlov, I. P., Pikielny, C. W., Frost, H. R., Her, S. C., Cole, M. D., Strohbehn, S. D., Wallace-Bradley, D., Kimmel, M., Gorlova, O. Y., & Amos, C. I. (2018). Gene characteristics predicting missense, nonsense and frameshift mutations in tumor samples. *BMC Bioinformatics*, *19*(1), 430. https://doi.org/10.1186/S12859-018-2455-0

[11] Smith, R. P., Lam, E. T., Markova, S., Yee, S. W., & Ahituv, N. (2012). Pharmacogene regulatory elements: From discovery to applications. *Genome Medicine*, *4*(5). https://doi.org/10.1186/GM344